D4.2 Stability and change

Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/sl-d4-2.html

Written for this site from the class slides, with textbook page references for each statement. The chapter is on pages 798–813 of the course textbook. Image credits are on the last slide of the slides for this subtopic.

Keeping ecosystems stable

D4.2.1 Stability in natural ecosystems

pp. 798–799

Forests, deserts and other ecosystems have continued over long periods, and some may have lasted for millions of years: rainforests in South East Asia are thought to be at least 70 million years old. This is **stability**.

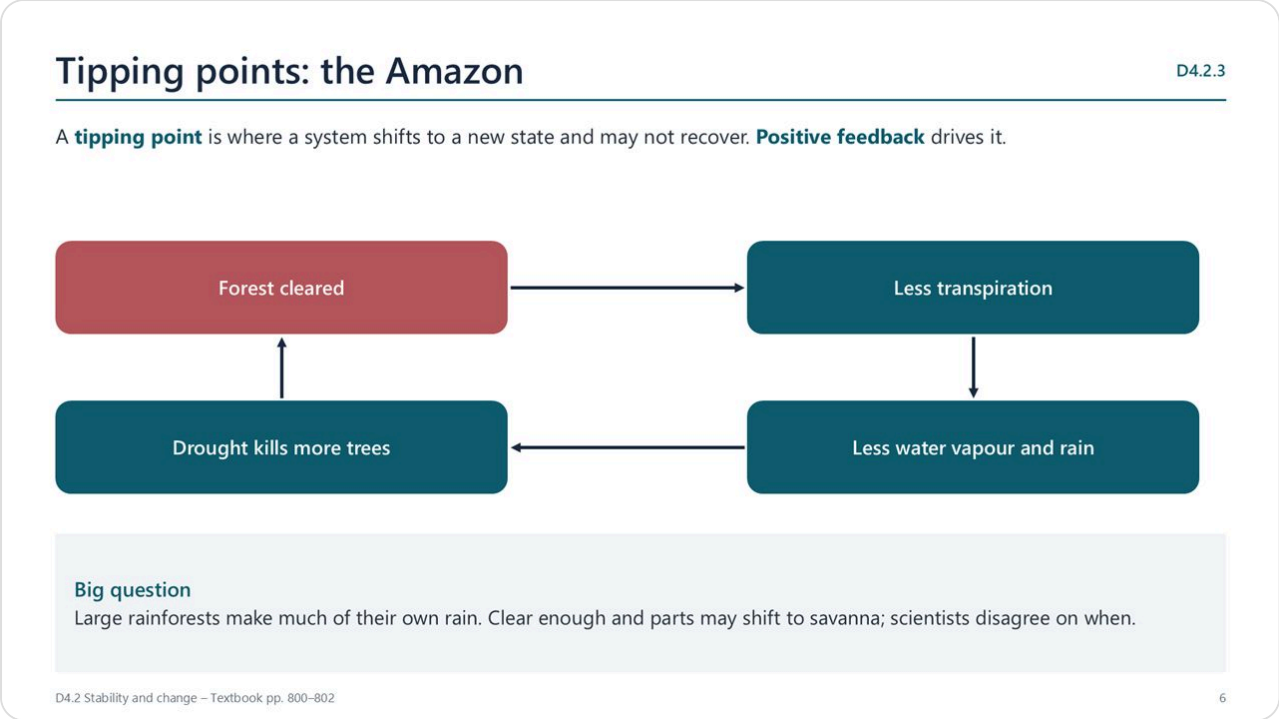
D4.2.2 What stability requires

pp. 799–803

Ecosystems need a supply of energy, recycling of nutrients, genetic diversity, and climate within tolerance limits. Sealed **mesocosms** can be used to investigate this, as energy can pass through, but matter cannot.

D4.2.3 The Amazon as a possible tipping point

pp. 800–802



A large area of rainforest is needed to produce atmospheric water vapour by transpiration, which brings cooling, air flows and rainfall. Removing trees reduces rain, which can kill more trees. A **tipping point** may be reached, beyond which the forest changes to another state. Some estimates put it at 20–25% deforestation, and about 17% of the original forest had gone by 2018.

D4.2.4 Keystone species

p. 805

These have a disproportionate effect on community structure, and the ecosystem may collapse if they are removed. Beavers and wolves are examples. So is the agouti, a rodent that buries Brazil nuts and so spreads the seeds of the Brazil nut tree.

Using natural resources

D4.2.5 Sustainability of harvesting

pp. 806–807

Harvesting is sustainable if its rate is lower than the rate of replacement. Timber in Finland is managed with a legal duty to regrow, whereas Atlantic cod stocks have collapsed from over-fishing.

D4.2.6 Sustainability of agriculture

pp. 807–808

Consider soil erosion, leaching of nutrients, the supply of fertilizers and other inputs, pollution from agrochemicals, and carbon footprint.

Pollution and recovery

D4.2.7 Eutrophication

pp. 808–810

Nitrate and phosphate from fertilizers leach into water, and algae bloom. Plants die in the shade, and decomposers use up oxygen, raising the **biochemical oxygen demand**. BOD is measured by keeping a water sample in the dark at 20 °C for five days and recording the fall in oxygen. Fish and other organisms can die as a result.

D4.2.8 Biomagnification

pp. 810–811

Toxins that build up in tissues reach higher concentrations at each trophic level, since each consumer eats many organisms. DDT and mercury are examples.

Watch out: bioaccumulation is build-up within one organism. Biomagnification is the rise from level to level.

D4.2.9 Plastics in the oceans

pp. 812–813

Plastics persist, because they are not biodegradable. **Macroplastics** entangle animals or fill their stomachs, as in Laysan albatross chicks and sea turtles. **Microplastics**, smaller than 5 mm, enter food chains.

D4.2.10 Rewilding

p. 813

Methods include returning apex predators and other keystone species, reconnecting habitats over large areas, and reducing human impact. Natural processes can then run again.

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